

Derrick Agyapong

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SUMMARY

IT professional and M.Sc. Computer Science student with a track record of partnering with business stakeholders to translate operational needs into working technology solutions. Experienced in requirements gathering, dashboard and reporting development, process improvement, and system implementation across research, healthcare, and enterprise environments. Background spans full-stack and backend development, database design and management, Agile/Scrum delivery, DevOps and cloud infrastructure (AWS), and applied data analysis and machine learning, with consistent experience communicating between technical teams and non-technical stakeholders.

EDUCATION

Master of Science, Computer Science

Aug 2024 – Dec 2026

University of North Dakota, Grand Forks, ND | GPA: 3.9

Bachelor of Science, Information Technology Management

Sep 2016 – May 2020

University of Professional Studies Accra (UPSA), Accra, Ghana | GPA: 3.4

EXPERIENCE

Graduate Research Assistant — Center for Cyber Security Research, UND

Sep 2024 – Present

Grand Forks, ND

- Partnered with subject matter experts and an industry utility partner to translate an operational security requirement into a real-time substation monitoring solution, funded under a U.S. Department of Energy CESER grant.
- Built a dashboard integrating acoustic, radar, camera, and bullet-trajectory triangulation data sources into a single interface for real-time monitoring and reporting.
- Pretrained a Vision Transformer (YOLOv8m) model to classify people, drones, wildlife, and vehicles for real-time site monitoring.
- Applied NIST cybersecurity frameworks (CSF, RMF, SP 800-series) and conducted machine-learning-based research for anomaly, threat, and intrusion detection in cyber-physical systems.
- Led large-scale data collection, preprocessing, and feature engineering; findings accepted for publication at the 2025 IEEE Cyber Awareness and Research Symposium.
- Conduct machine learning engineering and cybersecurity research, developing AI-driven solutions for threat detection, classification, and intelligent security systems.
- Develop technical specifications, system workflows, architecture designs, and documentation to support solution implementation.
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Graduate Research Assistant — School of Public Health, UND

Jun 2025 – Sep 2025

Grand Forks, ND

- Partnered with the North Dakota Perinatal Quality Collaborative (NDPQC) to gather and document requirements for a patient- and family-facing mobile solution.
- Designed and implemented a Flutter-based mobile application with location-based search, enabling users to find nearby healthcare and counseling facilities.
- Iterated on application enhancements and testing with stakeholders to ensure usability for non-technical end users.
- Conduct system analysis and identify opportunities to improve business processes through automation and technology.
- Analyze business and technical requirements and translate them into scalable, secure, and maintainable technology solutions.

System Administrator — Union of Professional Nurses and Midwives

Jan 2022 – Jul 2024

Accra, Ghana

- Served as the primary bridge between organizational leadership and IT systems, managing internal IT operations end to end.
- Developed the organization's website, internal application software, and web portals used daily by employees.
- Designed, built, and managed underlying databases and integrated RESTful APIs to support new functionality.

- Organized and led training sessions for employees and external stakeholders to drive adoption of new systems and processes.
- Oversee database architecture, API integration, application development, and cloud-based services supporting organizational operations.
- Design and implement end-to-end software solutions spanning web applications, APIs, databases, and cloud infrastructure.

Full-Stack Developer / DevOps Engineer / Scrum Master — Tiquiat Software

Jan 2020 – Dec 2021

Remote, United Kingdom

- Served as Scrum Master, running sprint planning, stand-ups, and retrospectives, and communicating project progress and priorities between stakeholders and the development team.
- Owned CI/CD pipeline deployment and cloud infrastructure on AWS (EC2, Step Functions, EventBridge, Amplify, SQS, SES, OpenSearch).
- Developed and integrated RESTful and GraphQL APIs; built and maintained full-stack web applications.
- Designed Entity Relationship Diagrams (ERD) and Context/Container/Component-level architecture diagrams to communicate system design to technical and non-technical audiences.

Backend Developer — Motes Systems

May 2018 – Oct 2021

Accra, Ghana

- Communicated regularly with stakeholders and cross-functional teams to align on integration requirements, application performance, and release priorities.
- Developed and integrated APIs with the frontend and managed CI/CD pipeline deployment.
- Wrote and maintained software test suites and used cloud platforms to deploy, monitor, and maintain applications, feeding reliability findings back to the team to drive process improvements.
- Work with cross-functional teams to troubleshoot complex technical issues and deliver solutions within project requirements and timelines.

PUBLICATIONS

- Agyapong, D., Jangula, J., Evan, D., Mohammed, F., & Ranganathan, P. (2025). Classification of Shockwave and Muzzle Blast Signatures Using CNN Models. 2025 Cyber Awareness and Research Symposium (CARS), IEEE.
- Agyapong, D., Sapkota, A., & Ranganathan, P. (2025). Spectrogram-Based Gunshot Detection Using Machine Learning Model. 2025 Cyber Awareness and Research Symposium (CARS), IEEE.
- Fiadzeawu, D. P., Agyapong, D., Ranganathan, P., & Zhang, J. (2025). Anomaly Detection in CPS using LSTM-Autoencoder with OCSVM on SWaT Dataset. 2025 Cyber Awareness and Research Symposium (CARS), IEEE.
- Alornyo, S. Y., Agyapong, D., Kibiwott, J. K., & Kim, E. (2025). Real-time Analysis of DDoS Attack Detection and Mitigation Measures Using XGBoost Machine Learning Algorithm. 2025 Cyber Awareness and Research Symposium (CARS), IEEE.

SKILLS

Business Analysis & Delivery: Requirements gathering & documentation, stakeholder management, Agile/Scrum, reporting & dashboards, process improvement, change management, user training & documentation, project management.

Databases: MySQL, PostgreSQL, MongoDB, Microsoft SQL Server, AWS DynamoDB, Supabase, Redis, SQL.

Programming & Web: Python, JavaScript/TypeScript, C++, PHP, Node.js, React, React Native, Nest.js, Express.js, Flutter.

Cloud & DevOps: AWS (EC2, Lambda, SageMaker, S3, EventBridge, Amplify, SQS, SES), Docker, Kubernetes, CI/CD, GitHub Actions, Git.

Data & Machine Learning: Data analysis & preprocessing, data visualization, feature engineering, TensorFlow, PyTorch, CNN, LSTM, Transformers, RAG, prompt engineering.

Cybersecurity: NIST CSF/RMF/SP 800-series, Nmap, Wireshark, Snort3, Burp Suite, OWASP ZAP, penetration testing.

General Tools: Microsoft Office Suite (Word, PowerPoint, Excel, Access, Outlook), Draw.io.